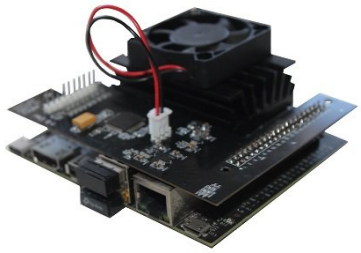
PCB & MANUFACTURING SPECIFICATION · SKU CQA240229JN

Board Material & Fabrication Data

Bare-board specification, mechanical features and the production quality test for JNEEG (board revision V2.0). Circuit design files are proprietary and are not part of this document.



FR-4	2	1.6 mm	79 × 100	0.25 mm
Base material	Copper layers	Board thickness	mm · L-shape	Minimum track

01 BARE-BOARD FABRICATION PARAMETERS

Parameters for ordering the bare PCB, as used by common PCB order forms (Elecrow, JLCPCB, PCBWay).

Parameter	Specification	Note
Base material	**FR-4**, standard Tg (130–140 °C)	Glass-epoxy laminate
Layers	**2** (top and bottom copper)	
Dimensions	**79.0 × 100.0 mm**	L-shaped outline with an internal corner; routed profile
Thickness	**1.6 mm**	Standard
Solder mask colour	**Black**	As on current production boards (earlier batches blue)
Silkscreen	**White**	Pin numbers, E1–E8, Ref, Biasout, warnings
Surface finish	**ENIG** recommended	Flat pads for the 0.5 mm-pitch TQFP-64; lead-free HASL acceptable
Copper weight	**1 oz** (35 µm) outer layers	
Minimum track / spacing	0.25 mm (10 mil)	Standard capability
Minimum hole	Ø0.49 mm	Vias
RoHS	Lead-free materials and process	

02 LAYER STACK-UP

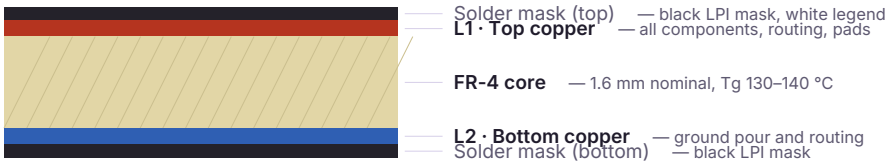


Figure 1 — 2-layer, 1.6 mm build. Layer roles are described in general terms only.

03 MECHANICAL FEATURES

Feature	Size	Qty	Purpose
Mounting holes	Ø2.70 mm	3	M2.5 stand-offs
40-pin socket holes	Ø1.10 mm plated	40	2 × 20 elevated socket, 2.54 mm
Electrode header holes	Ø1.00 mm plated	10	1 × 10 electrode header, 2.54 mm
Test header holes	Ø1.00 mm plated	4	J1 supply test socket
Vias	Ø0.49 mm	81	Layer transitions

For research and education only. Not a medical device. Specifications subject to change.

04 ASSEMBLY

Technology	Single-sided SMT (all parts on the top side) plus through-hole connectors
Key IC	TI ADS1299 (TQFP-64, 0.5 mm pitch) with ultra-low-noise on-board supplies
Passives	0402 and 0805 chip parts; one tantalum bulk capacitor
Connectors	2 × 20 elevated socket; 1 × 10 electrode header; 4-pin test socket
Indicators	2 green LEDs
Soldering	Lead-free reflow for SMD; selective or hand soldering for THT; no-clean flux, keep the analog input area free of residue
Marking	“JNEEG V2.0”, “power supply only from battery”, “not medical device”, pieeg.com

05 PRODUCTION QUALITY TEST

1 Supplies

Mount on a Jetson Nano powered from a battery. **LED D1** must light; J1 must read about −2.5 V, +2.5 V and 3.2 V against GND.

2 Data

Run `1.Graph_real_time.py` or `pieeg-server --device jneeg`. All 8 channels must show a live trace at 250 SPS.

3 Shorted-input check

Short **Ref** to each channel (or set CHnSET = 0x01). The channel must stay flat within a few µV.

06 HANDLING, STORAGE AND COMPLIANCE

Topic	Requirement
ESD and storage	ESD-sensitive: handle by the edges on an anti-static surface. Store dry in the anti-static bag, 10–35 °C.
Environment	Indoor use only. No conformal coating; keep away from liquids and conductive gel.
Classification	Development and evaluation board for research and education. Not a medical device.
Design files	Schematic, layout, Gerbers and BOM are proprietary to PiEEG and are not published.

07 RELATED DOCUMENTS

JN-DS-001
Product Datasheet

JN-MD-002
Mechanical & Dimensional Drawing

JN-PIN-004
Connector Pinout & Electrode Guide

JN-QSG-005
Quick Start & Safety Guide

JN-SW-006
Data Interface & Software Guide

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